TABLE OF CONSTANTS

Speed of light in free space	$c = 3.00 \times 10^{8} ms^{-1}$
Permeability of free space	$\mu_o = 4\pi \times 10^{-7} Hm^{-1}$
Permittivity of free space	$\varepsilon_o = 8.85 \times 10^{-12} Fm^{-1}$
	$= \left(\frac{1}{36\pi}\right) \times 10^{-9} Fm^{-1}$
Elementary charge	$e = 1.60 \times 10^{-19} C$
Planck's constant	$h = 6.63 \times 10^{-34} Js$
Unified atomic mass constant	$u = 1.66 \times 10^{-27} Kg$
Rest mass of electron	$m_e = 9.11 \times 10^{-31} Kg$
Rest mass of proton	$m_p = 1.67 \times 10^{-27} Kg$
Molar gas of constant	$R = 8.31 JK^{-1} mol^{-1}$
The Avogadro's constant	$N_A = 6.02 \times 10^{23} mol^{-1}$
The Boltzmann's constant	$k = 1.38 \times 10^{-23} JK^{-1}$
Gravitational constant	$G = 6.67 \times 10^{-11} Nm^2 Kg^{-2}$
Acceleration of free fall	$g = 9.81 ms^{-2}$

SECTION A: MULTIPLE CHOICE QUESTIONS

Answer ALL Questions

1. The dimensions of Momentum, Surface tension and pressure are respectively.

A. $ML^{-1}T^{-2}, ML^{2}T^{-3}$ & MLT^{-2}

B. MLT^{-1}, MT^{-2} & $ML^{-1}T^{-2}$

C. $ML^{2}T^{-2}, MLT^{-2}$ & MLT^{-1}

D. $ML^{-1}T^{2}, ML^{-2}T^{2}$ & MLT^{-2}

2. Given three vectors, $A = 3i - j + 3k, B = -I + 2j - k$ and $C = -3i + 5j - 3k$. Find; $(A + B).C$

A. -7

B. -8

C. -10

D. -14

3. When a box rests on the floor, it is acted upon by:

A. Gravitational and centrifugal forces

B. Gravitational force and acceleration

C. Gravitational and normal forces

D. Gravitational and frictional forces

4. If the net force applied in the direction of motion to a certain object on a horizontal frictionless surface is doubled, the acceleration of the object is?

A. Halved

B. Doubled

C. Unchanged

D. Quadrupled

5. A student weighing 700N climbs at constant speed to the top of an 8m vertical rope in $10s$.The average power expended by the student to overcome gravity is

$[g = 10ms^{-2}]$

A. $87.50W$

B. $560.0W$

C. $875.0W$

D. $5600W$

6. A projectile launched at 45° with a velocity of $100m/s$ hits a target. What will be the velocity if the range is doubled?

A. $141.4\ m/s$

B. $241.56\ m/s$

C. $41.5\ m/s$

D. $400.5\ m/s$

7. What is the instantaneous speed of a car moving according to the equation $x = (2t^3 + 4t^2 + 110t)\ at\ time\ t = 4secs$

A. $560\ m/s$

B. $238\ m/s$

C. $406\ m/s$

D. $450\ m/s$

8. A mass of $10g$ is suspended from the end of a spring whose force constant is $0.2N/m$. The mass is set into a simple harmonic motion. Calculate the period of the motion

A. $7.72s$

B. $11.41s$

C. $3.41s$

D. $3.41s$

9. Specific latent heat of evaporation of a liquid is

A. The hidden heat that changes the liquid to steam at its boiling point.

B. The heat that raises the temperature of a liquid by one degree at boiling point

C. The heat that converts the liquid at its boiling point to steam

D. The heat that changes a unit mass of the liquid to vapor at the same temperature

10. A swimming pool is filled with water of refractive index 1.33. Calculate the depth of the pool if the difference between the apparent and the real depth is $1.5m$.

A. $3.0m$

B. $4.5m$

C. $6.0m$

D. $7.5m$

11. A plane progressive wave is represented by $y = 3Sin\pi\ (200t + 7.5x)$, where the symbols have their usual meanings. The wavelength and speed are respectively

A. $1.75m, 57m/s$

B. $0.57m, 100m/s$

C. $0.27m, 27m/s$

D. $1.75m, 100m/s$

12. A student is far sighted with a near point of $100cm$. what power of lens will the student use to read a newspaper at a normal near point?

A. $33.3D$

B. $0.33D$

C. $3.0D$

D. $3.0D$

13. The measurement of a thermodynamic property known as temperature is based on

A. Zeroth law of thermodynamics

B. Second law of thermodynamics

C. First law of thermodynamics

D. Law of conservation of energy

14. The angle of incidence of a light ray from air to another medium is $30°$, while it was refracted in the second medium at an angle of $22°$. What is the refractive index in the second medium?

A. 1.42

B. 1.52

C. 1.35

D. 1.33

15. Calculate the mass of ice that would melt when $2kg$ of copper at the boiling point of water is quickly transferred to a block of ice at $0°C$ without heat loss.

$$\left[\begin{matrix} Specific\ heat\ capacity\ of\ copper = 400Jkg^{-1}K^{-1}, \\ Latent\ heat\ of\ fusion\ of\ ice = 3.3 \times 10^{5}Jkg^{-1} \end{matrix}\right]$$

A. $\frac{33}{80}kg$

B. $\frac{80}{33}kg$

C. $\frac{33}{8}kg$

D. $\frac{8}{33}kg$

16. The force of repulsion between two point charges $5\mu C$ and $8\mu C$ at a distance of $0.02m$ apart is $\left[\frac{1}{4\pi\varepsilon_o} = 9.0 \times 10^{9}Nm^{2}C^{-2}\right]$

A. $450N$

B. $900N$

C. $300N$

D. $1800N$

17. A battery of e.m.f of $24V$ and internal resistance 4Ω is connected to a resistor of 32Ω. What is the terminal potential?

A. $6V$

B. $24V$

C. $21.3V$

D. $16V$

18. Calculate the peak alternating current which can cause a power dissipation of $200W$ heating coil 1000Ω

A. $0.3A$

B. $0.63A$

C. $0.41A$

D. $1.0A$

19. The study of charges at rest as well as the laws that govern them is known as

A. Electrostatics

B. Electrodynamics

C. Electrophorus

D. Electromagnetism

20. Which of the following are true about electric lines of force?

I. They terminate from positive charges

II. They terminate on negative charges

III. They intersect each other

IV. They repel each other

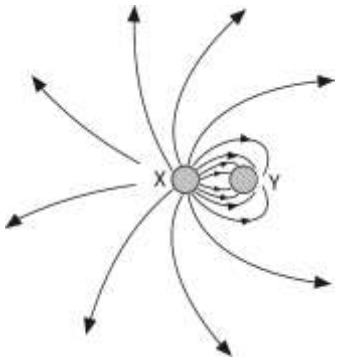
A. I, II, III and IV

B. I and II only

C. I, II and IV only

D. I and III only

21. Which of the following statements is correct about the charges X and Y in the figure below?



A. X is positive and Y is negative

B. X is negative and Y is positive

C. X and Y are both positive

D. X and Y are both negative

22. In which of the following processes is the knowledge of electrostatics applicable?

A. Washing of tiled floors.

B. Reduction of fly ash emissions

C. Driving of cars

D. Reduction of friction

23. The direction of force on a current-carrying conductor placed in a magnetic field can be found by

A. Cork screw rule

B. Fleming's left hand rule

C. Fleming's right hand rule

D. Using a compass

24. A semiconductor diode is used in rectifying alternating current into direct current mainly because it

A. Allows current to flow in either direction

B. Is non-linear

C. Offers a high input resistance

D. Allows current to flow in only one direction

25. If the bulk modulus of water is $8 \times 10^{11}\ N/m^2$. Estimate the change in volume of $300cm^3$ of water subjected to a pressure of $35mPa$.

A. $0.00013cm^3$

B. $0.0013cm^3$

C. $0.013cm^3$

D. $0.13cm^3$

26. Which of the following is true for a perfectly inelastic collision?

I. Kinetic energy is conserved

II. Coefficient of restitution is less than zero

III. Momentum is not conserved

IV. After collision, the particles travel with a common velocity

A. I and II only

B. III and IV only

C. II only

D. IV only

27. A laser beam is incident at an angle of $30.0°$ from the vertical onto a solution of corn syrup in water. If the beam is refracted to $19.24°$ from the vertical, what is the index of refraction of the syrup solution?

A. 1.52

B. 1.33

C. 1.44

D. 1.00

28. What is the internal energy of $1.0mol$ of an ideal monatomic atomic gas at $273K$?

$[R = 8.31\ J/molK]$

A. $3.2 \times 10^3 J$

B. $3.3 \times 10^3 J$

C. $3.4 \times 10^3 J$

D. $3.5 \times 10^3 J$

29. The gravitational force between two masses m_1 and m_2 which are at a distance r apart is given by $F = G\frac{m_1 m_2}{r^2}$ The dimensionality of the universal gravitational constant, G is

A. $M^{-1}L^3T^{-2}$

B. $M^3L^{-1}T^{-2}$

C. $ML^{-1}T^{-2}$

D. M^3LT^{-2}

30. A Carnot engine operates between temperature of $500K$ and $300K$ and accepts a heat input of $1200J$. The work output is

A. $720J$

B. $240J$

C. $480J$

D. $500J$

31. Two identical travelling waves, $y_1 = y_m \sin(kx - wt)$ and $y_2 = y_m$ are combined. What is the resultant amplitude produced by the two waves when destructive interference occurs?

A. 0

B. $\frac{1}{2}y_m$

C. y_m

D. $2y_m$

32. What is the correct order of increasing frequency and decreasing wavelength of the electromagnetic spectrum?

A. Radio wave→ Infra ray→ visible light →Ultraviolet light →X-ray→ Gamma ray

B. Gamma ray → X-ray → Ultraviolet → Visible light → Infra ray → Radio wave

C. Infra ray → Ultraviolet → X-ray → Radio wave → Visible light → Gamma ray

D. Visible light→ Gamma ray → Infra ray → Radio wave → X-ray → Ultraviolet

33. Example of a non-contact force is

A. Normal force

B. Frictional force

C. Magnetic force

D. Tensional force

34. A $20\mu F$ capacitor is connected to a $120V, 60Hz$ source. What is the capacitive reactance?

A. 132.6Ω

B. 177.6Ω

C. 265.5Ω

D. 123.5Ω

35. What amount of energy is stored in an inductor of inductance $1.5H$, when a current of $1.5A$ flows through it?

A. $6.25J$

B. $3.375J$

C. $1.69J$

D. $2.75J$

36. A radioactive material has a half-life of $24\ days$, its decay constant is

A. $2.880 \times 10^{-2}/day$

B. $0.693\ /day$

C. $1.44 \times 10^{-2}/day$

D. $3.38 \times 10^{-2}/day$

37. A spaceship at $20{,}000m$ height is orbiting the moon. If its motion is only affected by the gravitational pull of the moon, find the time it takes for a single orbit.
$[M_{moon} = 7.34 \times 10^{22} kg \; and \; R_{moon} = 1.738 \times 10^{6} \; m]$

A. $76 \; mins$

B. $96 \; mins$

C. $101 \; mins$

D. $110 \; mins$

38. Which of the following is true of an equipotential surface?

A. It is always at right angles to the lines of force

B. It is always parallel to the lines of force

C. It varies with the shape of the conductor

D. It can be drawn in the absence of an electric field

39. Which of the following expression is FALSE about photoelectric effect if h is the Planck's constant, f_0 is the threshold frequency for the metal, f is the frequency of the incident photon and φ is the work function for a given metal?

A. The maximum kinetic energy E_{Kmax} of an ejected electron is given by $E_{Kmax} = hf - \varphi$

B. The maximum kinetic energy E_{Kmax} of an ejected electron is thus $E_{Kmax} = h(f - f_0)$

C. The work function is given by $\varphi = hf_0$

D. The maximum kinetic energy E_{Kmax} of an ejected electron is $E_{Kmax} = h(f + f_0)$

40. Calculate the binding energy per nucleon for $^{14}_{28}Si$ which has a mass number of $27.97698amu$, given thatmass of a proton is $1.007825amu$ and the mass of neutron is $1.008665amu$

A. $8.54 \; MeV$

B. $8.64 \; MeV$

C. $8.44 \; MeV$

D. $8.43 \; MeV$

41. Which of the following transitions in a hydrogen atom emits the photon of lowest frequency?

A. $n_i = 4 \rightarrow n_f = 3$

B. $n_i = 3 \rightarrow n_f = 1$

C. $n_i = 4 \rightarrow n_f = 2$

D. $n_i = 2 \rightarrow n_f = 1$

42. From Rutherford's experiment on the structure of the atom, which of the following conclusion was made?

A. The atom is a sphere of positive charges embedded with negative charges

B. Electrons constitute most of the mass of the atom

C. The atom's positive charges were concentrated in a tiny, massive central core

D. The atom does not contain a nucleus

43. Caesium has a work function of $3 \times 10^{-19}\,J$. The maximum energy of liberated electrons when it is illuminated by light of frequency $6.7 \times 10^{14} Hz\ is$

$[h = 6.6 \times 10^{-34} Js]$

A. $1.42 \times 10^{-19} J$

B. $3.00 \times 10^{-19} J$

C. $4.42 \times 10^{-19} J$

D. $7.42 \times 10^{-19} J$

44. Which of the following is a wrong assumption of the Galilean transformation?

A. Time is the same in all frames

B. Length is different when measured from another frame of reference

C. A frame can move with constant relative velocity to another

D. Speed of light is constant

45. To break a chemical bond in the molecules of human skin and thus cause sun burn, photon energy of about $3.50eV$ is required. What is the wavelength of this photon?.

A. $354nm$

B. $451nm$

C. $875nm$

D. $762nm$

46. The most stable state of an electron in an atom is the

A. Excited state

B. $n = 0\ state$

C. Ground state

D. $n = \infty\ state$

47. The function of the $X - rays$ in the Millikan's oil drop experiment can be achieved using

A. Electric field

B. Magnetic field

C. Betatron

D. Radioactive material

48. Newton's laws of motion can be expressed in terms of momentum as

I. In the absence of external forces, total momentum of a given system remain constant

II. The rate of change of momentum is equal to the net applied force

III. Whenever two bodies exert forces on one another, the resulting changes on momentum are equal and opposite.

A. I, II and III

B. II only

C. III only

D. II and III only

49. Which of the following gives the correct relationship between the intensity of light, I and its amplitude, A?

A. $I \propto A^2$

B. $I \propto \sqrt{A}$

C. $I \propto \frac{1}{A^2}$

D. $I \propto \frac{1}{A^3}$

50. A work of $30J$ is done in transferring $5mC$ of charge from a point B to a point A in an electric field. The potential difference between B and A is

A. $1.5Kv$

B. $0.2Kv$

C. $3Kv$

D. $6Kv$

SECTION B: ESSAY QUESTIONS

Answer FOUR Questions in all; Not more than ONE Question from each course

PHY 001: MECHANICS AND PROPERTIES OF MATTER

1.a Give two examples each of derived and fundamental quantities. $(2\ marks)$

1.b Obtain from the first principle of the dimensions of the following quantities: acceleration, power, momentum and pressure. $(4\ marks)$

1.c The velocity of waves on an ocean of density ρ is written as $v = Ag^x\rho^y\lambda^z$, where A, x, y, z are constants, g is the acceleration due to the gravity, and λ the wavelength of the wave. Using dimensional analysis, find the exact values of $x, y\ and\ z$. $(4\ marks)$

2.a

i) Distinguish between the term turbulent and streamline flows $(2\ marks)$

ii) Define viscosity and mention two factors affecting viscosity in fluid $(2\ marks)$

2.b The hydraulic brake system of a car is such that its brake pedal is linked with the wheel's brake shoe by a suitable brake fluid. If the driver applies a force of $70N$ on a pedal which has an area of $2.00m^2$ at its point of contact with the fluid, calculate the force on the brake shoe of area $6.00m^2$ $(2\ marks)$

2.c

i) Differentiate between Conservative and Non-conservative forces. $(2\ marks)$

ii) A 5 kg block hangs from a string which passes over a frictionless pulley. The other end of the string is attached to a 12kg block that lies on a frictionless horizontal table. Calculate the acceleration of the two blocks and the tension in the string. $(2\ marks)$

PHY 002: HEAT, WAVES AND OPTICS

3.a

i) Differentiate between Constructive and destructive interference.

$(2\ marks)$

ii) State Boyle's law and Charles' laws writing their mathematical expressions.

$(3\ marks)$

3.b A $500g$ piece of a substance is heated to $100°C$ in boiling water and quickly dropped into $20g$ 0f water at $10°C$ contained in a calorimeter of mass $200g$. The final temperature of the mixture is $65.5°C$. Calculate the specific heat capacity of the substance neglecting heat lost to the surrounding.

$[Specific\ heat\ capacity\ of\ water = 4200\ Jkg^{-1}k^{-1}]$

$[specific\ heat\ capacity\ of\ calorimeter = 400\ Jkg^{-1}k^{-1}]$

$(3\ marks)$

3.c Briefly explain Isothermal change and Adiabatic change.

$(2\ marks)$

4.a The objective and eyepiece focal lengths of a compound microscope are $0.5cm$ and $12.5cm$ respectively. The objective lens forms a real image which is $15.0cm$ from the lens. Calculate the total magnification if the eye is held close to the eyepiece and views the virtual image at a distance of $25.0cm$.

$(3\ marks)$

4.b The image obtained with a converging lens is erect and four times the length of the object. The focal length of the lens is $20.0cm$. Calculate the image and object distances. $(4\ marks)$

4.b What lens is required to correct the defect in a defective eye whose far point is $150.0cm$? Also, determine the near point of the person. $(3\ marks)$

PHY 003: ELECTRICITY AND MAGNETISM

5.a

i) Define electric potential and electric field strength at a point in an electrostatic field. $(2\ marks)$

ii) A piece of wire of cross-sectional area $2mm^2$ has a resistance of 300Ω. Find the resistance of the wire of same length and material, if the cross-sectional area is $5mm^2$ $(3\ marks)$

5.b A cell supplies a current of $0.8A$ through a 2Ω resistor and $0.3A$ through a 7Ω resistor. What is the e.m.f and internal resistance of the cell? $(3\ marks)$

5.c State Faraday's law of electromagnetic induction and list three factors that affect the magnitude of the induced e.m.f.

$(2\ marks)$

6.a An electron accelerates in a uniform electric field of intensity $270{,}000\ N/C$. Calculate:

i) the acceleration of the electron $(1\ mark)$

ii) the force of the electron $(1\ mark)$

iii) the time it takes to travel $30mm$ from rest $(2\ marks)$

$[Charge\ of\ electron\ e\ = 1.6\ \times 10^{-19}C, mass\ of\ electron\ =\ 9.1 \times 10^{-31}kg]$

6.b An electron $m_e =\ 9.1\ \times 10^{-31}kg$ is placed between two parallel plate conductors separated by a distance of $25cm$. If the final velocity of the electron is $108m/s$, calculate:

i) the work done on the electron $(2\ marks)$

ii) the potential difference between the two plates. $(2\ marks)$

iii) the electric field between the plates. $(2\ marks)$

PHY 004: MODERN PHYSICS

7.a

i) State Bragg's law mathematically $(2\ marks)$

ii) X-rays are incident normally on a crystal lattice at $0.6\ nm$. The first order reflection occurs at an angle $30°$. What is the wavelength of X-rays used?

$(2\ marks)$

7.b List four properties of X-rays. $(2\ marks)$

7.c

i) What is an electron volt? $(1\ mark)$

ii) A metal has a work function $3.0eV$ illuminated by light of wavelength $0.6nm$. Calculate the:

(a) threshold frequency

(b) maximum kinetic energy of photo electrons in electron volt

$[h = 6.63 \times 10^{-34}Js, C = 3.0 \times 10^{8}m/s, e = 1.6 \times 10^{-19}C]$

$(3\ marks)$

8.a What is the difference between Compton effect and Photoelectric effect?

$(2\ marks)$

8.b The radius of the Sun is $6.96 \times 10^{8}m$, and its total power output is $3.85 \times 1026W$. Assuming the sun's surface emits on a black body, calculate:

i) Its surface temperature. $(2\ marks)$

ii) The maximum wavelength of light from the Sun

$[\sigma = 5.67 \times 10^{-8}Wm^{-2}K^{-4}, c = 2.898 \times 10^{-3}mK]$

$(2\ marks)$

8.c

i) What does Einstein's special theory of relativity state about the laws of Physics.

$(1\ mark)$

ii) What are radioisotopes? $(1\ mark)$

iii) State two (2) differences between nuclear fission and nuclear fusion.

$(2\ marks)$

SECTION A: MULTIPLE CHOICE ANSWERS

1. Option B

2. No Option

$A = 3i - j + 3k,$

$B = -i + 2j - k \; and$

$C = -3i + 5j - 3k$

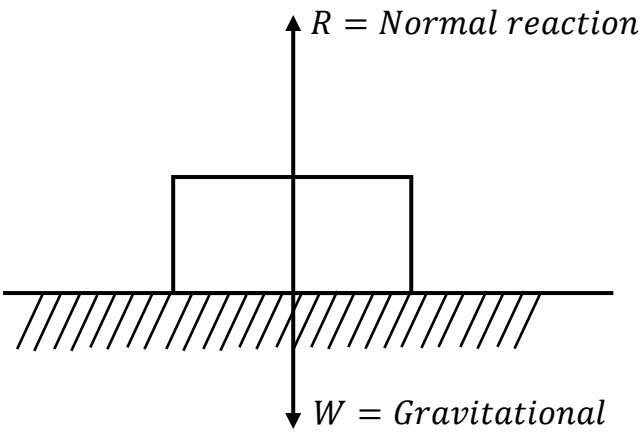
$A + B = (3i - j + 3k) + (-i + 2j - k)$

$A + B = 2i + j + 2k$

$(A + B) \cdot C = (2i + j + 2k) \cdot (-3i + 5j - 3k)$

$= -6 + 5 - 6 = -7$

3. Option C



4. Option B

$Using\ Net\ force, F = ma$

$F \propto a\ where\ mass, m\ is\ the\ constant\ of\ proportionality$

$F = Ka$

$K = \frac{F}{a}$

$\frac{F_1}{a_1} = \frac{F_2}{a_2}$, $\frac{F_1}{a_1} = \frac{2F_1}{a_2}$

$a2 = \frac{2F_1 a_1}{F_1}$

$\therefore\ a_2 = 2a_1$

The acceleration is doubled

5. Option B

$W = 700N, h = 8m, t = 10s, g = 9.8ms^{-2}$

Using $W = mg$

$m = \frac{w}{g} = \frac{700}{9.8} = 71.4kg$

$Power\ expended = \frac{Energy\ expended}{Time}$

$\therefore\ P = \frac{mgh}{t} = \frac{71.4 \times 9.8 \times 8}{10} = 560.0W$

6. Option A

$\theta = 45°, U_1 = 100ms^{-1}, U_2 =?$

Let the initial and final ranges be R_1 *and* R_2 *respectively*

$Using\ R = \frac{U^2\ Sin\ 2\theta}{g}$

$R = \frac{U^2\ Sin\ (45° \times 2)}{g} = \frac{U^2\ Sin\ 90°}{g} = \frac{U^2}{g}$

$U^2 = gR$

$U^2 \propto R$ *where* g *is the constant of proportionality*

$U^2 = KR$

$K = \frac{U^2}{R}$

$\frac{U_1^2}{R_1} = \frac{U_2^2}{R_2}$

$$\frac{100^2}{R_1} = \frac{U_2^2}{2R_1}$$

$$10000 = \frac{U_2^2}{R_1}$$

$$U_2^2 = 20000$$

$$U_2 = \sqrt{20000} = 141.4ms^{-1}$$

7. Option B

$$x = 2t^3 + 4t^2 + 110t$$

$$Instantaneous\ speed, v = \frac{dx}{dt}$$

$$v = 6t^2 + 8t + 110$$

$$At\ t = 4s$$

$$v = \frac{dx}{dt}\ where, t = 4, \qquad \frac{dx}{dt} = 6(4)^2 + 8(4) + 110$$

$$v = 6(16) + 32 + 110$$

$$v = 96 + 32 + 110 = 238ms^{-1}$$

8. Option D

$$m = 10g = 10 \times 10^{-3}kg = 0.01kg$$

$$K = 0.2Nm^{-1}, T = ?$$

$$Using\ T = 2\pi\sqrt{\frac{m}{K}}$$

$$T = 2 \times 3.142\sqrt{\frac{0.01}{0.2}}$$

$$T = 6.284\sqrt{0.05}$$

$$T = 6.284 \times 0.2236 = 1.40s$$

9. Option D

10. Option C

$a\eta_w = 1.33$

$Real\ depth - Apparent\ depth = 1.5$

$R - A = 1.5 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (i)$

$Refractive\ index, a\eta_w = \frac{R}{A}$

$\frac{R}{A} = 1.33$

$A = \frac{R}{1.33} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (ii)$

$Substitute\ (ii)\ into\ (i)$

$R - \frac{R}{1.33} = 1.5$

$\left(1 - \frac{1}{1.33}\right)R = 1.5$

$(1 - 0.75)R = 1.5$

$0.25R = 1.5$

$R = \frac{1.5}{0.25} = 6m$

11. Option C

$y = 3 \sin \pi(200t + 7.5x)$

$Compared\ with\ the\ equation\ of\ the\ form$

$y = A \sin \pi\left(2ft + \frac{2x}{\lambda}\right)$

$2f = 200, f = \frac{200}{2} = 100Hz$

$\frac{2}{\lambda} = 7.5$

$7.5\lambda = 2, \quad \lambda = \frac{2}{7.5} = 0.27m$

$Using\ v = f\lambda$

$v = 100 \times 0.27 = 27ms^{-1}$

12. Option D

For the student to see object clearly at $25cm$ from the eye, the image must appear to be at $100cm$.

$Hence, u = 25cm, v = -100cm$

$Using\ \frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

$\frac{1}{f} = \frac{1}{25} + \frac{1}{(-100)} = \frac{4 - 1}{100}$

$\frac{1}{f} = \frac{3}{100}$

$Power\ of\ the\ lens$

$P = \frac{1}{f} = \frac{3}{100 \times 10^{-2}m} = 3.0D$

13. Option A

14. Option D

$i = 30°, r = 22°$

$Refractive\ index\ in\ the\ second\ medium$

$a\eta_w = \frac{Sin\ i}{Sin\ r} = \frac{Sin\ 30}{Sin\ 22} = \frac{0.5000}{0.3746} = 1.33$

15. Option B

$Mass\ of\ Copper, M_c = 2kg, Initial\ temperature\ of\ Copper, \theta_c = 100°C$

$Let\ Mass\ of\ ice = m_i$

$Initial\ temperature\ of\ ice, \theta_i = 0°C$

$Specific\ heat\ capacity\ of\ Copper,\ C_c = 400Jkg^{-1}k^{-1}$

$Latent\ heat\ of\ fusion\ of\ ice,$

$Li = 3.3 \times 105Jkg^{-1}$

$Heat\ lost\ by\ Copper = M_c\ C_c\ (\theta_c - 0)$

$= 2 \times 400\ (100 - 0) = 800 \times 100$

$= 80000J$

$\therefore\ Heat\ lost\ by\ Copper = 8 \times 104J$

$Heat\ gained\ by\ ice\ =\ m_i\ \times\ 3.3\ \times\ 105$

$\therefore\ Heat\ gained\ by\ ice\ =\ 33\ \times\ 104m_i\ J$

$Provided\ no\ heat\ loss$

$Heat\ lost\ by\ Copper\ =\ Heat\ gained\ by\ ice\ 3.3\ \times\ 105m_i\ =\ 8\ \times\ 10^4$

$m_i\ =\ \frac{8\ \times\ 10^4}{33\ \times\ 10^4}\ =\frac{8}{33}Kg$

16. Option B

$Using\ Force\ of\ repulsion,$

$F\ =\ \frac{1}{4\pi\varepsilon_0}.\frac{q_1\ q_2}{r^2}$

$Where\ q_1\ =\ 5\mu C\ =\ 5\ \times\ 10^{-6}C$

$q_2\ =\ 8\mu c\ =\ 8\ \times\ 10^{-6}C, r\ =\ 0.02m$

$F\ =\ \frac{9\ \times 10^9\ \times\ 5\ \times\ 10^{-6}\ \times\ 8\ \times 10^{-6}}{(0.02)^2}\ =\ \frac{0.36}{0.0004}$

$=\ 900N$

17. Option C

$E\ =\ 24v, r\ =\ 4\Omega, R\ =\ 32\Omega, I\ =?$

$Using\ I\ =\ \frac{E}{R\ +\ r}\ =\ \frac{24}{32+4}\ =\ \frac{24}{36}$

$Terminal\ potential$

$=\ P.d\ across\ external\ resistance$

$V\ =\ IR\ =\ 0.67\ \times\ 32\ =\ 21.3v$

18. Option B

$P\ =\ 200W, R\ =\ 1000\Omega, I\ =?$

$Using\ P\ =\ I^2R$

$I^2\ =\ \frac{P}{R}$

$I\ =\ \sqrt{\frac{P}{R}}\ =\ \sqrt{\frac{200}{1000}}\ =\sqrt{0.2}\ =\ 0.45A$

$\therefore\ I_{rms} = 0.45A$

$I_{rms} = \frac{I_0}{\sqrt{2}}$

$Peak\ current,\ I_0 = \sqrt{2}I_{rms}$

$I_0 = \sqrt{2} \times 0.45$

$I_0 = 1.41 \times 0.45 = 0.63A$

19. Option A

20. Option C

21. Option A

22. Option D

23. Option D

24. Option C

25. Option C

$B = 8 \times 10^{11} N/m^2$

$V = 300cm^3 = 300 \times (10^{-2})^{-3} m^2 = 300 \times 10^{-2} m^3$

$P = 35mPa = 35 \times 10^{-3} N/m^2$

$Using\ Bulk\ Modulus$

$B = \frac{PV}{\Delta V}$

$\Delta V = \frac{PV}{B}$

$\Delta V = \frac{35 \times 10^{-3} \times 300 \times 10^{-6}}{8 \times 10^{11}} = \frac{1.05 \times 10^{-5}}{8 \times 10^{11}}$

$\Delta V = 1.3 \times 10^{-3} m^3 = 1.3 \times 10^{-17} \times (10^2)^3 cm^3$

$\Delta V = 1.3 \times 10^{-17} \times 10^6 cm^3 = 1.3 \times 10^{-11} cm^3$

$\Delta V = 0.013 \times 10^{-9} cm^3$

$\therefore \Delta V = 0.013ncm^3$

26. Option D

27. Option A

$Angle\ of\ incident, i\ =\ 30°$

$Angle\ of\ refraction, r\ =\ 19.24°$

$Applying\ Snell's\ law$

$Index\ of\ refraction,$

$$\eta\ =\ \frac{Sin\ i}{Sin\ r}\ =\ \frac{Sin\ 30}{Sin\ 19.24}\ =\ \frac{0.5000}{0.3295}=\ 1.52$$

28. Option C

$No\ of\ mole, n\ =\ 1mole$

$Temperature, T\ =\ 273K$

$Molar\ gas\ constant, R\ =\ 8.31J/molK$

$Using\ Internal\ energy$

$U\ =\ nC_vRT$

$For\ monoatomic\ gas, C_v\ =\ 3/2$

$$U\ =\ 1\ \times\ \frac{3}{2}\ \times\ 8.31\ \times\ 273\ =\ 3403J$$

$$=\ 3.4\ \times\ 10^3J$$

29. Option A

$If\ F\ =\ G\frac{m_1m_2}{r^2}$

$$G\ =\ \frac{Fr^2}{m_1m_2}$$

$The\ dimension\ of;\ F\ =\ MLT^{-2}, r^2\ =\ L^2$

$m_1m_2\ =\ M^2$

$$G\ =\ \frac{MLT^{-2}\ \times\ L^2}{M^2}\ =\ M^{-1}L^3T^{-2}$$

30. Option C

$T_H\ =\ 500K, T_C\ =\ 300K, Q\ =\ 1200J, W\ =?$

$$Using\ \frac{T_C}{T_H}\ =\ \frac{Q_C}{Q_H}$$

$$\frac{300}{500} = \frac{Q_C}{1200} = 500Q_C = 360000$$

$$Q_C = \frac{360000}{500} = 720J$$

$$Workdone, W = Q_H - Q_C$$

$$W = 1200 - 720 = 480J$$

31. Option A

$$y_1 = y_m \sin(kx - wt) \; and$$

$$y_2 = y_m \sin(kx - wt + \pi)$$

$$y = y_1 + y_2$$

$$y_m \sin(kx - wt) + y_m \sin(kx - wt + \pi)$$

$$y = y_m[\sin(kx - wt) + \sin(kx - wt + \pi)]$$

$$Recall \; that;$$

$$\sin A + \sin B = 2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$$

$$Let \; A = kx - wt \; and \; B = kx - wt + \pi$$

$$y = 2y_m \sin\left[\frac{(kx - wt) + (kx - wt + \pi)}{2}\right]\cos\left[\frac{(kx - wt) - (kx - wt + \pi)}{2}\right]$$

$$y = 2y_m \sin\left[\frac{kx - wt + kx - wt + \pi}{2}\right]\cos\left[\frac{kx - wt - kx - wt + \pi}{2}\right]$$

$$y = 2y_m \sin\left[\frac{\pi - 2wt}{2}\right]\cos\left[\frac{-\pi}{2}\right]$$

$$y = 2y_m \sin\left[\frac{\pi}{2} - wt\right]\cos\left[\frac{-\pi}{2}\right]$$

$$y = 2y_m \sin\left[\frac{\pi}{2} - wt\right] \times 0$$

$$y = 0$$

32. Option A

33. Option C

34. Option A

$C = 20\mu F = 20 \times 10^{-6}F, V = 120V,$

$f = 60Hz$

$Using\ capacitive\ reactance,$

$$X_C = \frac{1}{2\pi fc}$$

$$X_C = \frac{1}{2 \times 3.142 \times 60 \times 20 \times 10^{-6}} = \frac{1}{7.5408 \times 10^{-3}}$$

$$X_C = 132.6\Omega$$

35. Option C

$L = 1.5H, I = 1.5A$

$Using\ Energy\ stored\ in\ an\ inductor$

$$E = \frac{1}{2} I^2 L$$

$$E = \frac{1}{2}(1.5^2)(1.5)$$

$$E = \frac{1}{2}(2.25)(1.5)$$

$$E = \frac{1}{2}(3.375) = 1.69J$$

36. Option A

$Half-life, \quad T_{\frac{1}{2}} = 24\ days$

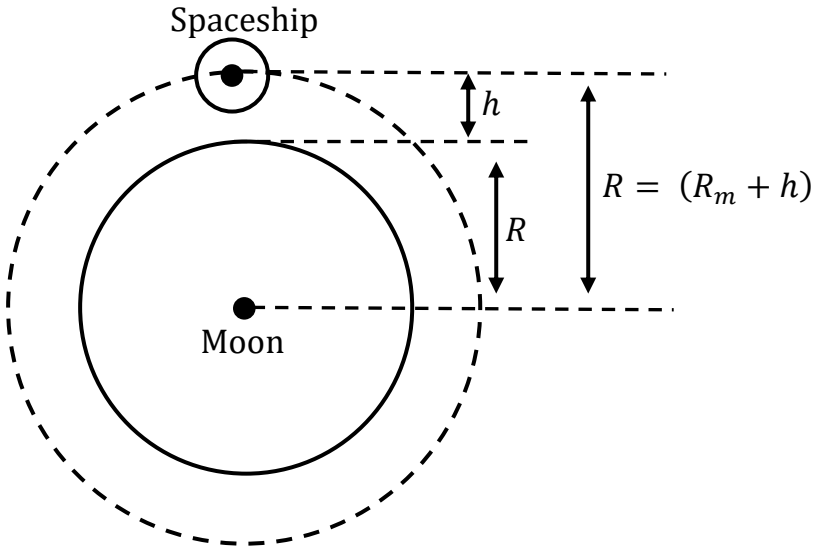
$Decay\ constant, \lambda = ?$

$$Using\ T_{\frac{1}{2}} = \frac{0.693}{\lambda}$$

$$\lambda = \frac{0.693}{T_{\frac{1}{2}}} = \frac{0.693}{24} = 0.0288/day$$

$$\therefore \lambda = 2.88 \times 10^{-2}/day$$

37. Option B



$h = 20{,}000m = 2 \times 104 = 0.02 \times 106m$

$M_{moon} = 7.34 \times 1022kg$

$R_{moon} = 1.73 \times 106m$

$Radius\ of\ the\ orbit, R = R_{moon} + h$

$R = 1.73 \times 106 + 0.02 \times 106 = 1.75 \times 106m$

$Using\ Time\ taken\ for\ a\ single\ orbit$

$$T = 2\pi\sqrt{\frac{R^3}{g_{moon}R_{moon}^2}}$$

$But\ g_{moon}R_{moon}^2 = GM_{moon}$

$$Then\ T = 2\pi\sqrt{\frac{R^3}{GM_{moon}}}$$

$$T = 2 \times 3.142\sqrt{\frac{5.359375 \times 10^{18}}{(6.7 \times 10^{-11}) \times (7.34 \times 10^{22})}}$$

$$T = 6.284\sqrt{\frac{5.359375 \times 10^{18}}{4.9178 \times 10^{12}}}$$

$$T = 6.284\sqrt{1089791.167}$$

$$T = 6.284 \times 1043.93 = 6560.06s$$

$$T = \frac{6560.06}{60} = 110mins$$

38. Option A

39. Option D

40. Option C

$Mass\ of\ Silicon\ nucleus = 27.97698amu$

$Mass\ of\ Proton = 1.007825amu$

$Mass\ of\ neutron = 1.008665amu$

$^{14}_{28}Si + Energy = 14\ neutrons + 14\ protons$

$Energy = 14\ (neutrons + protons) - {}^{14}_{28}Si$

$Binding\ energy = 14(1.008665 + 1.007825) - 27.97698$

$E_B = 14(2.01649) - 27.97698$

$E_B = 28.23086 - 27.97698 = 0.25388amu$

$1amu = 931Mev$

$E_B = 0.25388 \times 931 = 236.36Mev$

$$Binding\ energy\ per\ nucleon = \frac{E_B}{28\ nucleons}$$

$$Binding\ energy\ per\ nucleon = \frac{236.36}{28}$$

$= 8.44Mev$

41. Option D

42. Option C

43. Option A

44. Option A

$E = 3.50eV$

$E = 3.50 \times (1.6 \times 10^{-19})$

$E = 5.6 \times 10^{-19}J$

$C = 3 \times 108ms^{-1}$

$$Using\ E\ =\ \frac{hc}{\lambda}$$

$$\lambda\ =\ \frac{hc}{E}$$

$$\lambda\ =\ \frac{6.6\ \times 10^{-34}\ \times\ 3\ \times\ 10^{8}}{5.6\ \times\ 10^{-19}}\ =\ \frac{1.9875\ \times\ 10^{-25}}{5.6\ \times\ 10^{-19}}$$

$$\lambda\ =\ 3.54\ \times 10^{-7}$$

$$\lambda\ = 354\ \times 10^{-9} m$$

$$\lambda\ =\ 354 nm$$

45. No Option

46. Option C

47. Option A

48. Option B

49. Option A

50. Option D

SECTION B: ESSAY ANSWERS

PHY 001: MECHANICS AND PROPERTIES OF MATTER

1.a

Examples of Fundamental quantities	**Examples of Derived quantities**
Mass	Velocity
Length	Acceleration
Time	Force
Current	Pressure
Temperature	Power

1.b

i) $Velocity\ (v) = \frac{Displacement}{Time}$

$V = \frac{L}{T} = LT^{-1}$

$Acceleration, a = \frac{v}{t} = \frac{LT^{-1}}{T}$

$a = LT^{-1} \times T^{-1} = LT^{-2}$

$\therefore Dimension\ of\ acceleration = LT^{-2}$

ii) $Power\ (P) = \frac{Energy(E)}{time(t)}$

$Energy(E) = Force\ (F) \times distance(s)$

$F = Mass \times acceleration$

$F = MLT^{-2}$

$E = MLT^{-2} \times L$

$E = ML^2T^{-2}$

$P = \frac{ML^2T^{-2}}{T} = ML^2T^{-2} \times T^{-1}$

$P = ML^2T^{-3}$

$\therefore Dimension\ of\ Power = ML^2T^{-3}$

iii) $Momentum\ (P)\ =\ Mass\ \times\ velocity\ (v)$

$v\ =\ LT^{-1}$

$P\ =\ MLT^{-1}$

$\therefore\ Dimension\ of\ momentum\ =\ MLT^{-1}$

iv) $Pressure\ =\ \frac{Force(F)}{Area(A)}$

$F\ =\ MLT^{-2}\ and\ A\ =\ L^2$

$P\ =\ \frac{MLT^{-2}}{L^2}\ =\ MLT^{-2}\times\ L^{-2}$

$\therefore\ Dimension\ of\ Pressure\ =\ ML^{-1}T^{-2}$

1.c $V\ =\ Ag^x\rho^y\lambda^z$

$Take\ dimension\ of\ both\ sides$

$LT^{-1}\ =\ A(LT^{-2})^x\ \times\ (ML^{-3})^yL^z$

$LT^{-1}\ =\ AL^xT^{-2x}M^yL^{-3y}L^z$

$M^0L^1T^{-1}\ =\ M^yL^{x\ -\ 3y\ +\ z}\ +\ T^{-2x}$

$Compared\ the\ indices\ of\ L.H.S\ with\ R.H.S$

$-\ 1\ =-\ 2x$

$x\ =\ \frac{-1}{-2}\ =\ \frac{1}{2}$

$y\ =\ 0$

$x\ -\ 3y\ +\ z\ =\ 1$

$Substitute\ the\ values\ of\ x\ and\ y\ into\ the\ equation$

$\frac{1}{2}\ -\ 3(0)\ +\ z\ =\ 1$

$\frac{1}{2}\ +\ z\ =\ 1$

$z\ =\ 1\ -\ \frac{1}{2}\ =\ \frac{1}{2}$

$\therefore\ \ x\ =\ \frac{1}{2}, y\ =\ 0\ and\ \ z\ =\ \frac{1}{2}$

2.a

i) Turbulent flow is a type of flow in which the path line of a fluid particle crosses and moves in a zig – zag or disorderly manner while stream line flow is a type of flow in which the path line do not cross one another and move along well defined path (orderly with manner).

(1) Turbulent flow is a flow with high Reynolds number and characterized by high flow rate while streamline is a flow with low Reynolds number and characterized by low flow rate.

(2) Streamline flow is laminar whereas turbulent flow is non – laminar

(3) Streamline flow is an orderly type of motion in which the fluid flows in parallel layers while in turbulent flow, the fluid flows in disorderly manner.

(4) In streamline flow, the velocity of the fluid is constant at any point in the fluid whereas in turbulent flow, the velocities of the particles of the fluid are different at any point in the fluid

ii) Viscosity is a measure of fluid resistance to flow or gradual deformation by shear stress or tensile stress. It describes the internal friction which exists between layers of a moving fluid.

Factors affecting viscosity in fluid are:

(1) Temperature

(2) Surface area

(3) Velocity

(4) Time

2.b $Force\ in\ the\ brake\ pedal, F_P = 70N$

$Cross - Sectional\ area\ of\ the\ brake\ pedal, A_P = 2.00m^2$

$Cross - Sectional\ area\ of\ the\ brake\ shoe, A_s = 6.00m^2$

$Let\ the\ force\ on\ the\ brake\ shoe\ be\ F_S$

Since pressure is transmitted equally in all directions,

pressure exerted on the brake pedal = Pressure exerted on the brake shoe

$$\frac{F_P}{A_P} = \frac{F_S}{A_s}$$

$$\frac{70}{2} = \frac{F_S}{6}$$

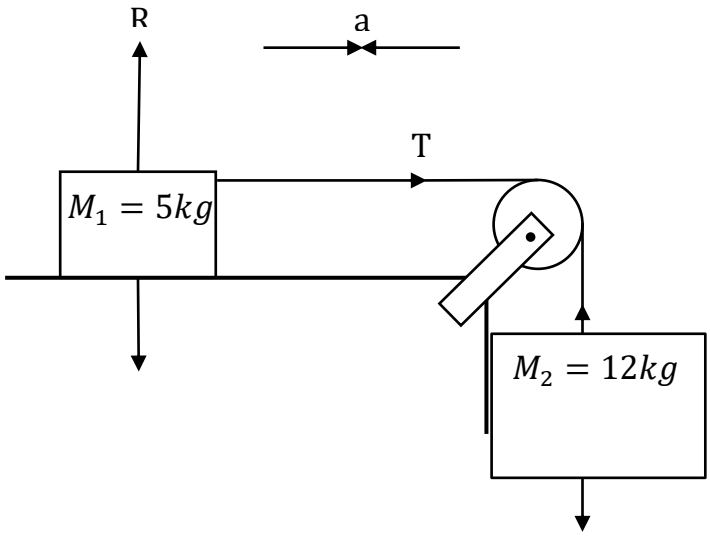
$$2F_S = 70 \times 6$$

$$F_S = \frac{420}{2} = 210N$$

2.c

i) Conservative forces are forces whose work done in a closed path equal to zero and are completely recoverable while Non – conservative forces are forces in which work done in a closed path is not equal to zero and are not recoverable.

ii)



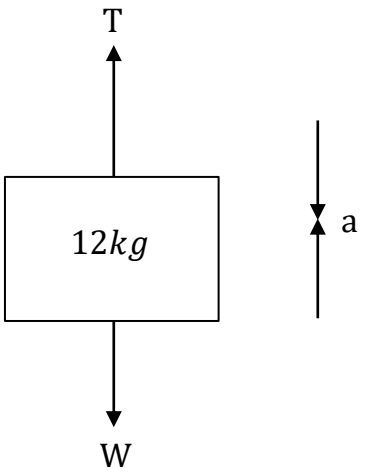
$For\ the\ mass, m_1\ moving\ along\ the\ plane,$

$Since\ the\ surface\ is\ frictionless, the\ frictional\ force, F_f = 0$

$Resultant\ force, R_f = T$

$m_1 a = T$

$5a = T$ …… (i)



$Resultant\ force, R_f = W_2 - T$

$m_2 g - T = ma$

$12a = (12 \times 9.8) - T$

$12a = 117.6 - T$ …… (ii)

$Add\ eqn\ (i)\ and\ (ii)$

$5a = T$ …… (i)

$12a = 117.6 - T$ …… (ii)

$17a = 117.6$

$a = \frac{117.6}{17} = 6.9ms^{-2}$

$Substitute\ the\ value\ of\ a\ into\ (i)$

$T = 5a$

$T = 5 \times 6.9 = 34.5N$

PHY 002: HEAT, WAVES AND OPTICS

3.a

i)

(1) Constructive interference occurs when the displacement of the incident waves that meet are in the same direction while destructive interference occurs when the displacement of the incident waves that meet are in the opposite direction.

(2) Constructive interference occurs when the amplitude of the resultant wave is larger than the amplitudes of incident waves. Therefore, the intensity of the resultant waves is larger than the intensity of incident waves while destructive interference occurs when the amplitude of the resultant wave is smaller than the amplitude of incident waves. Therefore, the intensity of the resultant waves is smaller than the intensity of incident waves.

ii) Boyle's law states that at constant temperature the volume of a fixed mass of a gas is inversely proportional to its pressure.

The Boyle's law can be expressed mathematically as;

$$V \propto \frac{1}{P} (T \; is \; constant)$$

$$V = \frac{K}{P} \; where \; K = constant$$

$$K = PV$$

$$If \; P_1, V_1 \; are \; the \; initial \; pressure \; and \; volume \; of \; a \; fixed \; mass \; of \; gas$$
$$and \; P_2, V_2 \; are \; the \; final \; pressure \; and \; volume, then: P_1V_1 = P_2V_2$$

Charles' law states that at constant pressure the volume of a fixed mass of gas is directly proportional to the absolute temperature.

The Charles' law can be expressed mathematically as;

$$V \propto T \; (P \; is \; constant)$$

$$V = KT \; where \; K \; is \; a \; constant$$

$$K = \frac{V}{T}$$

If V_1, T_1 are the initial volume and temperature of a fixed mass of gas, V_2, T_2 are the final volume and temperature, then: $\frac{V_1}{T_1} = \frac{V_2}{T_2}$

3.b Mass of Substance, $M_s = 500g = 0.5Kg$

Initial temperature of substance,

$\theta_s = 100°C$

Mass of water, $M_W = 20g = 0.02Kg$

Initial temperature of water and calorimeter, $\theta_{WC} = 10°C$

Mass of Calorimeter, $M_C = 200g = 0.2Kg$

Final temperature of mixture, $\theta_m = 65.5°C$

Specific heat capacity of water, $C_W = 4200JKg^{-1}K^{-1}$

Specific heat capacity of calorimeter, $C_c = 400JKg^{-1}K^{-1}$

Specific heat capacity of substance, C_s =?

Heat gained by water and calorimeter

$= M_W C_W(\theta_m - \theta_{WC}) + M_C C_C (\theta_m - \theta_{WC})$

$= (M_W C_W + M_C C_C)(\theta_m - \theta_{WC})$

$= ([0.02 \times 4200] + [0.2 \times 400])(65.5 - 10)$

$= (84 + 80)(55.5) = (164 \times 55.5)$

$= 9102J$

Heat lost by the substance

$= M_s C_s (\theta_s - \theta_m)$

$= 0.5\, C_s\, (100 - 65.5) = 0.5 \times 34.5\, C_s = 17.25C_s$

Neglect Heat loss to the surrounding,

Heat gained by water and calorimeter = Heat lost by substance

$17.25C_s = 9102$

$C_s = \frac{9102}{17.25} = 527.65JKg^{-1}K^{-1}$

3.c Isothermal change is a change of a system in which the temperature remains constant $i.e\ \Delta T\ =\ 0$

Adiabatic change is change in which a system exchanges no heat with its surroundings $i.e\ \Delta\theta\ =\ 0$

4.a $f_o\ =\ 0.5cm,\ f_e\ =\ 12.5cm, V_o\ =\ 15.0cm, D\ =\ 25.0cm$

$Using\ Magnification\ of\ compound\ microscope,$

$$M\ =\ \left(\frac{V_o}{f_o}-1\right)\left(\frac{D}{f_e}+1\right)$$

$$M\ =\ \left(\frac{15}{0.5}-1\right)\left(\frac{25}{12.5}+1\right)$$

$$M\ =\ (30\ -\ 1)(2\ +\ 1)$$

$$M\ =\ 29\ \times\ 3\ =\ 87$$

$Alternatively,$

$For\ the\ objective\ lens$

$u_o\ =\ ?, v_o\ =\ 15.0cm, f_o\ =\ 0.5$

$Applying\ lens\ formula$

$$\frac{1}{f_o}\ =\ \frac{1}{u_o}\ +\ \frac{1}{v_o}$$

$$\frac{1}{u_o}\ =\ \frac{1}{f_o}\ -\frac{1}{v_o}$$

$$\frac{1}{u_o}\ =\ \frac{1}{0.5}\ -\frac{1}{15}$$

$$\frac{1}{u_o}\ =\ \frac{30\ -\ 1}{15}\ =\ \frac{29}{15}$$

$$u_o\ =\ \frac{15}{29}cm$$

$Magnification\ produced\ by\ objective\ lens\ M_o\ =\ \frac{v_o}{u_o}\ =\ \frac{15}{15/29}$

$$M_o\ =\ 15\ \times\ \frac{29}{15}\ =\ 29$$

$For\ the\ eyepiece\ lens$

$u_e = ?, v_e = -25cm, f_e = 12.5cm$

Applying lens formula

$$\frac{1}{f_e} = \frac{1}{u_e} + \frac{1}{v_e}$$

$$\frac{1}{u_e} = \frac{1}{f_e} - \frac{1}{v_e}$$

$$\frac{1}{u_e} = \frac{1}{12.5} - \frac{1}{(-25)}$$

$$\frac{1}{u_e} = \frac{2 + 1}{25} = \frac{3}{25}$$

$$\therefore u_e = \frac{25}{3} cm$$

$$Magnification\ produced\ by\ eyepiece\ lens, M_e = \frac{v_e}{u_e} = \frac{25}{25/3}$$

$$M_e = 25 \times \frac{3}{25} = 3$$

$$Magnification\ of\ compound\ microscope, M = M_o \times M_e$$

$$M = 29 \times 3 = 87$$

4.b $focal\ length, f = 20.0cm$

Let the length of the object be u

Length of the image

$$v = 4u$$

Applying lens formula

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$\frac{1}{u} + \frac{1}{4u} = \frac{1}{20}$$

$$\frac{4 + 1}{4u} = \frac{1}{20}$$

$$\frac{5}{4u} = \frac{1}{20}$$

$$4u = 5 \times 20 = 100$$

$u = \frac{100}{4} = 25cm$

$The\ distance\ of\ the\ object = 25cm$

$v = 4u$

$v = 4 \times 25 = 100cm$

4.c The image of a distance object must be made to appear to be at $90cm$ in order to been seen Hence, $u = \infty\ (infinity)$

$v = -150.0cm$

$Using\ \frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

$\frac{1}{f} = \frac{1}{\infty} + \frac{1}{(-150)}$

$\frac{1}{f} = \frac{1}{(-150)}$

$f = -150cm$

A diverging lens of focal length $150cm$ is required to correct the defect for near point,

$u = -25cm, f = 150cm$

$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

$\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{150} - \frac{1}{25}$

$\frac{1}{v} = \frac{1 - 8}{150} = \frac{-7}{150}$

$\therefore v = \frac{-150}{7} = -21.4cm$

PHY 003: ELECTRICITY AND MAGNETISM

5.a

i) Electric potential at a point is defined as the work done in bringing unit positive charge from infinity to that point against the electricity force of the field. It is a scalar quantity and the S.I unit is $volt\ (V)$.

ii) Using $R = \frac{\rho l}{A}$

Since the wire is of the same length and material, L and ρ are constants of proportionality

$$R \propto \frac{1}{A}$$

$$R = \frac{K}{A}, K = RA$$

$$R_1A_1 = R_2A_2$$

$$Where\ R_1 = 300\Omega, R_2 = ?, A_1 = 2mm^2, A_2 = 5mm^2$$

$$300 \times 2 = R_2 \times 5$$

$$R_2 = \frac{600}{5} = 120\Omega$$

5.b $Using\ E = IR + Ir$

$$When\ I = 0.8A\ and\ R = 2\Omega$$

$$E = 0.8 \times 2 + 0.8r$$

$$E = 1.6 + 0.8r \ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (i)$$

$$When\ I = 0.3A\ and\ R = 7\Omega$$

$$E = (0.3 \times 7) + 0.3r$$

$$E = 2.1 + 0.3r \ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (ii)$$

$$Subtract\ eqn\ (ii)\ from\ (i)$$

$$0 = -0.5 + 0.5r$$

$$0.5r = 0.5$$

$$r = \frac{0.5}{0.5} = 1\Omega$$

$$Substitute\ the\ value\ of\ r\ into\ (i)$$

$E = 1.6 + 0.8r$

$E = 1.6 + 0.8(1)$

$E = 1.6 + 0.8 = 2.4V$

5.c Faraday's law of electromagnetic induction states that when there is a change in the magnetic line of force, the induced $emf\ (E)$ is directly proportional to the rate of change of the magnetic flux $(N\phi)$ linked with the circuit;

Mathematically,

$$E \propto \frac{\Delta N\phi}{\Delta t}, E = \frac{Kd\phi}{dt}$$

Where K = constant of proportionality

Factors that affect the magnitude of induced e.m.f are

(a) Magnetic strength of the core in the coil of wire

(b) Number of turns in the coil of wire

(c) Cross – sectional area of the coil

6.a $E = 270000\ N/C,\ M_e = 9.1 \times 10^{-31} Kg, q = 1.6 \times 10^{-19} C$

i) $E = \frac{F}{q}, F = Eq$

$m_e a = Eq$

$$a = \frac{Eq}{m_e} = \frac{2.7 \times 10^5 \times 1.6 \times 10^{-19}}{9.1 \times 10^{-31}}$$

$$a = \frac{4.32 \times 10^{-14}}{9.1 \times 10^{-31}}$$

$a = 4.75 \times 10^{16} ms^{-2}$

ii) $Using\ F = Eq$

$F = 2.7 \times 105 \times 1.6 \times 10^{-19}$

$F = 4.32 \times 10^{-14} N$

iii) $Distant\ travelled,$

$S = 30mm = 30 \times 10^{-3}m$

$S = 3 \times 10^{-2}m$

$u = 0ms^{-1}, a = 4.75 \times 10^{16}ms^{-2}$

$Using\ S = ut + \frac{1}{2}at^2$

$3 \times 10^{-2} = 0(t) + \frac{1}{2}(4.75 \times 10^{16})t^2$

$3 \times 10^{-2} = 2.375 \times 10^{16}t^2$

$t^2 = \frac{3 \times 10^{-2}}{2.375 \times 10^{16}} = 1.26 \times 10^{-18}$

$t = 1.26 \times 10^{-18} = 1.12 \times 10^{-9}sec$

$t = 1.12ns$

6.b

i) $m_e = 9.1 \times 10^{-31}Kg, d = 25cm = 0.25m, u = 0ms^{-1}, v = 10^8ms^{-1}$

$Workdone, W = Fd$

$But\ F = m_e a_w = m_e ad$

$Then\ V^2 = U^2 + 2ad$

$a = \frac{V^2 - U^2}{2d}$

$a = \frac{(10^8)^2 - 0^2}{2 \times 0.25}$

$a = \frac{10^{16}}{0.5} = 2.0 \times 10^{16}ms^{-2}$

$W = \frac{(9.1 \times 10^{-31}) \times (2.0 \times 10^{16})}{0.5} = 9.1 \times 10^{-15}J$

ii) $Using\ W = qV$

$V = \frac{q}{W} = \frac{1.6 \times 10^{-19}}{9.1 \times 10^{-15}} = 1.76 \times 10^{-5}V$

iii) $Using\ E = \frac{V}{d} = \frac{1.76 \times 10^{-5}}{0.25}$

$= 7.04 \times 10^{-5}NC^{-1}$

PHY 004: MODERN PHYSICS

7.a

i) $2d\sin\theta = n\lambda$

ii) $d = 0.6\,nm = 0.6 \times 10^{-9}m = 6 \times 10^{-10}m,$

$n = 1, \theta = 30°$

$Using\ 2d\sin\theta = n\lambda$

$2 \times 6 \times 10^{-10} \times \sin 30 = 1 \times \lambda$

$\lambda = 2 \times 6 \times 10^{-10} \times \sin 30 = 6 \times 10^{-10}m$

$\lambda = 6Å$

7.b The properties of x-rays are

(1) They travel with the same speed as that of visible light

(2) They cannot be reflected, refracted or deflected by a magnet or electric field as they do not possess any charge.

(3) They are pure energy, no mass and they transfer energy from place to place in the form of quanta (photons)

(4) They have shorter wavelength ranging from $10^{-2}nm\ to\ 10nm$.

(5) They have higher frequencies from $3 \times 10^{16}Hz\ to\ 3 \times 10^{19}Hz$.

7.c

i) An Electron volt is defined as the amount of energy $1.6 \times 10^{-19}J$ an electron gains after being accelerated by 1 $volt$ of electricity.

ii) Work Function, $\Phi = 3.0eV, \lambda = 0.6nm, \lambda = 0.6 \times 10^{-9}m = 6.0 \times 10^{-10}m$

(a) $Using\ Work\ function\ \Phi = hf_o$

$$f_o = \frac{\Phi}{h} = \frac{3 \times 1.6 \times 10^{-19}}{6.625 \times 10^{-34}} = \frac{4.8 \times 10^{-19}}{6.625 \times 10^{-34}}$$

$f_o = 7.25 \times 10^{14}Hz$

$Maximum\ kinetic\ energy\ of\ photo\ electrons,$

$E_{Kmax} = hf - \Phi$

$$E_{Kmax} = \frac{hc}{\lambda} - \Phi$$

$$E_{Kmax} = \frac{6.625 \times 10^{-34} \times 3 \times 10^{8}}{6 \times 10^{-10}} - 4.8 \times 10^{-19}$$

$$E_{Kmax} = \frac{1.99 \times 10^{-25}}{6 \times 10^{-10}} - 4.8 \times 10^{-19}$$

$$E_{Kmax} = 3.31 \times 10^{-16} - 4.8 \times 10^{-19}$$

$$E_{Kmax} = 3310 \times 10^{19} - 4.8 \times 10^{-19}$$

$$E_{Kmax} = (3310 - 4.8) \times 10^{-19} = 3305.2 \times 10^{-19} J$$

$$E_{Kmax} = \frac{3305.2 \times 10^{-19}}{1.6 \times 10^{-19}} = 2065.75 eV$$

$$\therefore E_{Kmax} = 2.06575 \times 10^{3} eV = 2.07 KeV$$

8.a In Compton Effect the photon $(x - rays\ or\ gamma\ rays)$ loses parts of it energy to the recoiled electrons, interacts with free electron and is scattered. Therefore, it takes place with free electron while in photo electric effect, the photon (visible light) is absorbed and loses its entire energy to the emitted electron and afterwards there is no photon. Therefore, it cannot take place with free electron, so it has to be a bound electron.

8.b $Radius\ of\ the\ Sun,\ R_S = 6.96 \times 10^{8} m,$

$Power, P = 3.85 \times 10^{26} W$

$\sigma = 5.67 \times 10^{-8} W m^{-2} k^{-4}, c = 2.898 \times 10^{-3} mk$

i) $Using\ Stefan's\ law, P = \varepsilon \sigma A T^{4}$

$$T^{4} = \frac{P}{\varepsilon \sigma A}, T = \sqrt[4]{\frac{P}{\varepsilon \sigma A}}$$

$For\ blackbody, \varepsilon = 1$

$Since\ Sun\ is\ spherical\ in\ shape, Surface\ area\ of\ the\ sun, A = 4\pi R_S^2$

$$A = 4 \times 3.142 \times (6.96 \times 10^{8})^2$$

$$A = 12.568 \times 4.84416 \times 1017$$

$$A = 6.088 \times 10^{18} m^2$$

ii) Differences between nuclear fission and nuclear fusion

Nuclear fission	**Nuclear fusion**
It involves the reaction in which a heavy nucleus is splitted into two lighter nuclei	It involves the reaction in which two lighter nuclei combined to create a heavy nucleus
It requires less amount of energy to split nucleus	It requires greater amount of energy to fuse nuclei
It releases tremendous amount of energy	It releases relatively high amount of energy
It is controllable	It is uncontrollable
Temperature is high	Temperature is extremely high